06/09/2026

CODE-F

Aakash
Medical | IIT-JEE | Foundations

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TERM EXAM FOR NEET-2027 (XII STUDYING) PH-I

MM: 720 **TE-03F** **Time: 180 Mins.**

Answers

1.	(4)	37.	(1)	73.	(4)	109.	(3)	145.	(4)
2.	(4)	38.	(4)	74.	(3)	110.	(1)	146.	(2)
3.	(3)	39.	(1)	75.	(2)	111.	(4)	147.	(1)
4.	(2)	40.	(4)	76.	(4)	112.	(3)	148.	(4)
5.	(1)	41.	(2)	77.	(4)	113.	(3)	149.	(2)
6.	(2)	42.	(4)	78.	(1)	114.	(1)	150.	(1)
7.	(2)	43.	(4)	79.	(4)	115.	(3)	151.	(1)
8.	(3)	44.	(2)	80.	(4)	116.	(4)	152.	(4)
9.	(3)	45.	(1)	81.	(1)	117.	(4)	153.	(3)
10.	(3)	46.	(1)	82.	(4)	118.	(4)	154.	(2)
11.	(3)	47.	(3)	83.	(2)	119.	(2)	155.	(2)
12.	(1)	48.	(1)	84.	(2)	120.	(4)	156.	(4)
13.	(2)	49.	(4)	85.	(3)	121.	(2)	157.	(4)
14.	(4)	50.	(4)	86.	(1)	122.	(1)	158.	(2)
15.	(1)	51.	(4)	87.	(3)	123.	(4)	159.	(1)
16.	(4)	52.	(3)	88.	(4)	124.	(1)	160.	(3)
17.	(3)	53.	(4)	89.	(4)	125.	(3)	161.	(1)
18.	(4)	54.	(1)	90.	(4)	126.	(3)	162.	(4)
19.	(2)	55.	(2)	91.	(4)	127.	(3)	163.	(2)
20.	(2)	56.	(1)	92.	(3)	128.	(3)	164.	(1)
21.	(3)	57.	(3)	93.	(4)	129.	(2)	165.	(3)
22.	(1)	58.	(2)	94.	(4)	130.	(1)	166.	(3)
23.	(4)	59.	(1)	95.	(1)	131.	(2)	167.	(2)
24.	(4)	60.	(1)	96.	(2)	132.	(4)	168.	(3)
25.	(4)	61.	(4)	97.	(1)	133.	(3)	169.	(1)
26.	(3)	62.	(3)	98.	(4)	134.	(1)	170.	(3)
27.	(1)	63.	(2)	99.	(1)	135.	(4)	171.	(1)
28.	(1)	64.	(4)	100.	(2)	136.	(4)	172.	(3)
29.	(1)	65.	(4)	101.	(3)	137.	(1)	173.	(1)
30.	(1)	66.	(3)	102.	(1)	138.	(3)	174.	(1)
31.	(2)	67.	(2)	103.	(3)	139.	(3)	175.	(1)
32.	(2)	68.	(2)	104.	(2)	140.	(3)	176.	(3)
33.	(2)	69.	(1)	105.	(2)	141.	(2)	177.	(2)
34.	(4)	70.	(3)	106.	(3)	142.	(3)	178.	(2)
35.	(1)	71.	(3)	107.	(1)	143.	(2)	179.	(2)
36.	(3)	72.	(1)	108.	(1)	144.	(3)	180.	(3)

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Hints and Solutions

PHYSICS

(1) **Answer :** (4)
Solution:
$B_0 = 510 \times 10^{-9}$ T, $E_0 = ?$
$C = \frac{E_0}{B_0}$
$E_0 = CB_0 = 3 \times 10^8 \times 5.1 \times 10^{-7} = 153$ N/C

(2) **Answer :** (4)
Solution:
Microwaves has largest wavelength among the following.

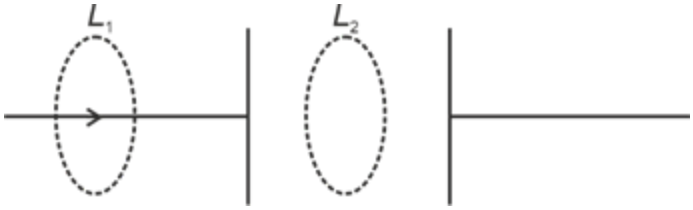
(3) **Answer :** (3)
Solution:
The direction of propagation of electromagnetic wave is in the direction of $\vec{E} \times \vec{B}$.

(4) **Answer :** (2)
Solution:
$i_d = \varepsilon_0 A \frac{dE}{dt} = \varepsilon_0 \pi R^2 \frac{dE}{dt} = i$
$i'_d = \varepsilon_0 \left(\pi R^2 - \frac{\pi R^2}{16}\right) \frac{dE}{dt}$
$= \frac{15}{16} \varepsilon_0 \pi R^2 \frac{dE}{dt} = \frac{15}{16} i$

(5) **Answer :** (1)
Solution:
$\frac{1}{2} \varepsilon_0 E^2 = (U)_{avg}$
$\Rightarrow (U)_{avg} = \frac{1}{2} \times \varepsilon_0 \times 4 = 2\varepsilon_0$
$= 17.7 \times 10^{-12}$ J/m^3

(6) **Answer :** (2)
Solution:
Velocity of wave in air, $C = \frac{1}{\sqrt{\mu_0 \varepsilon_0 \varepsilon_{r_1}}}$
Velocity of wave in medium, $v = \frac{1}{\sqrt{\mu_0 \varepsilon_0 \varepsilon_{r_1}}} = \frac{C}{3}$
$\Rightarrow \frac{1}{\varepsilon_{r2}} = \frac{1}{9} \times \frac{1}{\varepsilon_{r_1}} \Rightarrow \frac{\varepsilon_{r_1}}{\varepsilon_{r_2}} = \frac{1}{9}$

(7) **Answer :** (2)
Solution:



According to modified Ampere's law
$\oint B.\ dl = \mu_0 (I_C + I_D)$
for Loop L_1 $I_C \neq 0$ and $I_D = 0$
For Loop L_2 $I_C = 0$ and $I_D \neq 0$
Due to KCL $I_C = I_D$

(8) **Answer :** (3)
Solution:

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Gauss's law in electrostatics is

$\oint \vec{E} . \vec{ds} = \frac{q}{\varepsilon_0}$

Faraday's law is $\oint \vec{E} . \vec{dl} = -\frac{d\phi_B}{dt}$

Gauss's law in magnetism is $\oint \vec{B} . \vec{dA} = 0$

Ampere-Maxwell law is

$\oint \vec{B} . \vec{dl} = \left(\mu_0 i_c + \mu_0 \varepsilon_0 \frac{d\phi_E}{dt}\right)$

(9) **Answer :** (3)
Solution:

$\left(I = 1350 \frac{W}{m^2}\right)$

$P = \frac{2I}{C}$

$= \frac{2\times 1350}{3\times 10^8}$

= 9 × 10^{-6} Pa

(10) **Answer :** (3)
Solution:

$E_0 = C \times B_0 = 3 \times 10^8 \times 2 \times 10^{-7} = 60$ V/m, and the direction of electric field will be along *z*-axis

(11) **Answer :** (3)
Solution:

$\mu = \frac{C}{V} = \frac{\frac{1}{\sqrt{\mu_0 \varepsilon_0}}}{\frac{1}{\sqrt{\mu \varepsilon}}}$ = $\sqrt{\frac{\mu\varepsilon}{\mu_0\varepsilon_0}}$

(12) **Answer :** (1)
Solution:

$|\varepsilon_L| = L\frac{dI}{dt}$

$\Rightarrow\ L = \frac{1.6}{2.4}$

$\boxed{L = \frac{2}{3} H}$

(13) **Answer :** (2)
Solution:
In parallel combination of n equal inductor, Leq =L/n

(14) **Answer :** (4)
Solution:
induced electric field is non conservative and so it have closed field lines

(15) **Answer :** (1)
Solution:

$i = \left|\frac{1}{R}\left(\frac{d\phi}{dt}\right)\right|$

$i = \left|\frac{1}{10}\left(12t - 5\right)\right|$

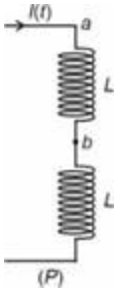
i = 0.2 A

(16) **Answer :** (4)
Solution:
By Fleming's right hand rule.

(17) **Answer :** (3)
Solution:
In figure (*P*)

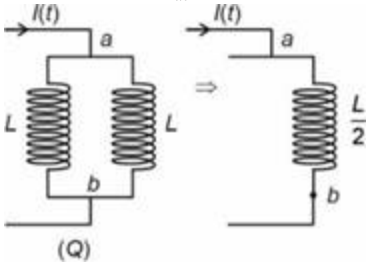
$E_P = E_{ab} = -\frac{Ldl(t)}{dt}$ ...(1)

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In figure (*Q*)

$E_Q = E_{ab} = -\frac{L}{2}\frac{dI(t)}{dt}$ ...(2)



$\Rightarrow \quad \frac{E_P}{E_Q} = \frac{2}{1}$

(18) **Answer :** (4)
Solution:
ε = $BL_{eff}v$ (L_{eff} = Diameter)
= $B2Rv$

(19) **Answer :** (2)
Solution:

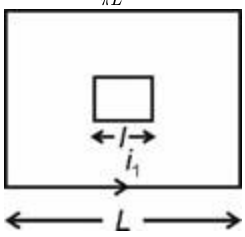
$d\varepsilon = \int_R^{2R} B\omega x dx$

$= \frac{B\omega}{2}\left[x^2\right]_R^{2R}$

$= \frac{3B\omega R^2}{2}$

(20) **Answer :** (2)
Solution:
Magnetic field at centre due to bigger coil

$B_1 = \frac{2\sqrt{2}\mu_0 i_1}{\pi L}$



$M = \frac{B_1 A_2}{i_1} = \frac{2\sqrt{2}\,\mu_0 l^2}{\pi L}$

Flux linked with smaller loop.

(21) **Answer :** (3)
Solution:

$\varepsilon = -N\frac{d\phi}{dt}$

$\left|\frac{\varepsilon}{R}\right| = \frac{N}{R}\frac{d\phi}{dt}$

$dq = \frac{N}{R}d\phi$

$\Delta Q = \frac{N(\Delta\phi)}{R}$

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$\Delta Q = \frac{\Delta\phi_{\text{total}}}{R}$

$= \frac{(NBA)}{R}$

$= \frac{\mu_0 ni\pi r^2}{R}$

Putting values

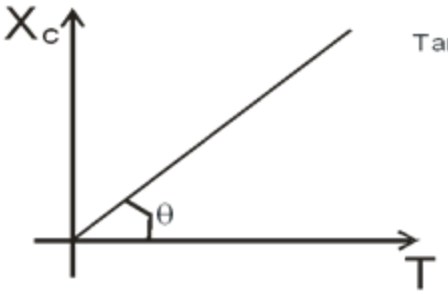
$= \frac{4\pi\times10^{-7}\times100\times4\times\pi\times(0.01)^2}{10\pi^2}$

$\Delta Q = 32\ \mu C$

(22) **Answer :** (1)
Solution:
Faradays laws are based upon conservation of energy.

(23) **Answer :** (4)
Solution:
$U = \frac{1}{2}L\,i^2 = \frac{1}{2}\times10\times10^{-3}\times1^2 = 5$ mJ

(24) **Answer :** (4)
Solution:



$\text{Tan}\theta = \frac{1}{2\pi C}$

$X_C = \frac{1}{\omega C} = \frac{1}{2\pi fC} = \frac{T}{2\pi C}$

$X_C \alpha T$

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(25) **Answer :** (4)
Solution:
The input voltage is 8 V DC = constant
ϕ_B
at primary and secondary is constant
E_{IND} =
$\frac{-d\phi_B}{dt} = 0$

(26) **Answer :** (3)
Solution:
$V = \sqrt{v_C^2 + v_R^2}$

$V = \sqrt{(30)^2 + (40)^2} = 50\,V$

(27) **Answer :** (1)
Solution:
Given circuit is in resonance. The average power dissipated in an AC circuit is given by the formula $\mathbf{P_{av} = I_{rms}^2 \times R =}$ 2500 W, . The rms value of current through the circuit is $i_{rms} = 50\sqrt{2}$ A .The phase difference between source emf and current is ϕ =0

(28) **Answer :** (1)
Solution:
$Z = \sqrt{R^2 + {X_L}^2}$

$Z = \sqrt{(40)^2 + (30)^2} = \sqrt{1600 + 900} = 50$ ohm

(29) **Answer :** (1)
Solution:
$I_1 = I_2 < I_3$ and $I_3 = I_3$; $I_1 = I_2 = \frac{I_0}{\sqrt{2}}$

(30) **Answer :** (1)
Solution:
Current will be maximum at the condition of resonance. So resonance angular frequency
$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{0.5 \times 8 \times 10^{-6}}} = 500$ rad/s

(31) **Answer :** (2)
Solution:
$V = 50 \times 2 \sin 100 \pi t \cos 100 \pi t$
$V = 50 \sin 200 \pi t$
$V_0 = 50$ volt, $f = 100$ Hz

(32) **Answer :** (2)
Solution:
$I_2 = \frac{I_1}{2}$
$\Rightarrow \frac{v}{z_2} = \frac{1}{2}\left(\frac{v}{z_1}\right)$
$z_2 = 2z_1$
$\sqrt{R^2 + X_2^2} = 2\sqrt{R^2 + X_1^2}$
$\Rightarrow R^2 + X_2^2 = 4\left(R^2 + X_1^2\right)$
$\Rightarrow R^2 + \left(\frac{1}{\omega_2 C}\right)^2 = 4\left[R^2 + \left(\frac{1}{\omega_1 C}\right)^2\right]$
$\Rightarrow R^2 + \left(\frac{3}{\omega_1 C}\right)^2 = 4\left[R^2 + \left(\frac{1}{\omega_1 C}\right)^2\right]$
$\Rightarrow R^2 + 9X_1^2 = 4R^2 + 4X_1^2$
$\Rightarrow 3R^2 = 5X_1^2$
$\Rightarrow \frac{X_1}{R} = \sqrt{\frac{3}{5}}$

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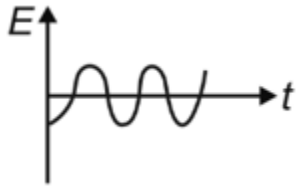
(33) **Answer :** (2)
Solution:
$\frac{L_1}{R} < \frac{L_2}{R}$
$L_1 < L_2$

(34) **Answer :** (4)
Solution:
$80 = \frac{P_{output}}{2\ kW}$
$P_{\text{output}} = 2 \times 10^3 \times \frac{80}{100} = 1600$ W
$P_{\text{output}} = I_S V_S$
$\frac{1600}{3.2} = V_S$
$V_S = 500$ V

(35) **Answer :** (1)
Solution:
For a pure inductive AC circuit:
$X_L \propto f$
For d.c. circuit
$f = 0$
$X_L = \omega L$
$= 2\pi f L$

$= 0$

(36) **Answer :** (3)
Solution:



$\phi = A\sin(\omega t)$

$\varepsilon = -\frac{d\phi}{dt} = -A\omega\cos\omega t$

(37) **Answer :** (1)
Solution:
Eddy current is induced when metal block is placed in changing magnetic field.

(38) **Answer :** (4)
Solution:

$\frac{\text{Energy}}{\text{Volume}} = \frac{B^2}{2\mu_0} \Rightarrow \text{Energy} = \frac{B^2(AL)}{2\mu_0}$

(39) **Answer :** (1)
Solution:
The average value over one half cycle would be zero as area covered would be zero

(40) **Answer :** (4)
Solution:
f = 10 Hz, V_{rms} = 12 V, C = 2.1 μF

$l_{\text{rms}} = \frac{V_{\text{rms}}}{X_C}, \quad X_C = \frac{1}{\omega C}$

ω = 2πf
Putting all values
$I_{\text{rms}} \cong$ 1.6 mA

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(41) **Answer :** (2)
Solution:

$\cos\phi = \frac{R}{Z} = 0$

(42) **Answer :** (4)
Solution:
At resonance $l = \frac{V}{R}$

$f = \frac{1}{2\pi\sqrt{LC}}$

And voltage across resistor, inductor and capacitor will be in different phase.

$I = \frac{200}{20}$

I = 10 A

$f = \frac{1}{2\pi\sqrt{\frac{2}{\pi}\times 10^{-6}\times\frac{2}{\pi}}}$

$= \frac{1000}{4}$

= 250 Hz

(43) **Answer :** (4)
Solution:
EM wave can transport both energy and momentum.

(44) **Answer :** (2)
Solution:
Displacement current $i_d = i$

(45) **Answer :** (1)
Solution:
($\omega t - kz$) indicates wave travels in +ve z direction.

CHEMISTRY

(46) **Answer :** (1)
Solution:

$[Co(NH_3)_4Cl_2]Cl + AgNO_3 \rightarrow AgCl$ (1 mole) + $[Co(NH_3)_4Cl_2]^+$

$[Co(NH_3)_3Cl_3] + AgNO_3 \rightarrow$ No precipitate of AgCl

$[Co(NH_3)_5Cl]Cl_2 + 2AgNO_3 \rightarrow 2AgCl + [Co(NH_3)_5Cl]^{+2}$

(47) **Answer :** (3)
Solution:

C.NO. = 6
O.S = 3

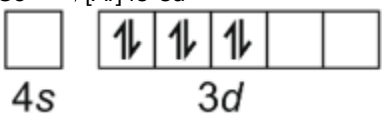
(48) **Answer :** (1)
Solution:
The complex with type $[Ma_3b_3]$ type of complexes show fac and mer type of isomers.
$\therefore$ $[Cr(py)_3(Cl)_3]$ shows fac and mer isomers.

(49) **Answer :** (4)
Solution:
The correct decreasing order of field strength according to crystal field theory is
$CO > NH_3 > H_2O > Cl^-$

(50) **Answer :** (4)
Solution:
Dimethyl glyoxime reagent is used as co-ordinating reagent in the quantitative estimation of nickel.

(51) **Answer :** (4)
Solution:
Vitamin B_{12} is the coordination compound of cobalt.

(52) **Answer :** (3)
Solution:
In $[Co(NH_3)_6]^{3+}$;
$Co^{3+} \Rightarrow [Ar]45°3d^6$



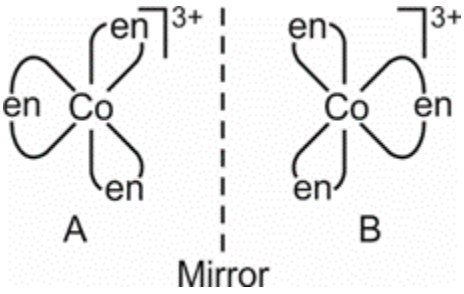
Here d^2sp^3 hybridisation occur.

(53) **Answer :** (4)
Solution:
$K_2[NiCl_4]$, Paramagnetic, sp^3 hybridised.
$\therefore$ Configuration = $e^4t_2^4$

(54) **Answer :** (1)
Solution:
Tetraammineaquachloridocobalt(III) chloride
$[Co(NH_3)_4(H_2O)Cl]Cl_2$

(55) **Answer :** (2)
Solution:

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A and B are enantiomers hence show optical isomerism.

(56) **Answer :** (1)
Solution:

M – C back bond is weaker in $[Mn(Co)_6]^+$ hence, C – O bond is strongest among the given species.

(57) **Answer :** (3)
Solution:
$[Rh(PPh_3)_3Cl]$ is Wilkinson's Catalyst.

(58) **Answer :** (2)
Solution:
Because of –CN it can show linkage isomerism

(59) **Answer :** (1)
Solution:
A. $[Co(NH_3)_5(NO_2)]Cl_2$ – Linkage isomerism due to 'N' and 'O' linkage by NO_2
B. $[Co(NH_3)_5(SO_4)]Br$ – Ionization isomerism
C. $[Co(NH_3)_6][Cr(CN)_6]$ – Coordination isomerism
D. $[Co(H_2O)_6]Cl_3$ – Solvate isomerism

(60) **Answer :** (1)
Solution:
Magnetic moment, $\mu = \sqrt{n(n+2)}\ BM$ (where n = number of unpaired electrons)

	Complex	No. of unpaired electron(s)	μ(BM)
a.	$[Fe(CN)_6]^{3-}$	1	1.73
b.	$[Fe(H_2O)_6]^{3+}$	5	5.92
c.	$[Fe(CN)_6]^{4-}$	0	0
d.	$[Fe(H_2O)_6]^{2+}$	4	4.90

(61) **Answer :** (4)
Solution:

A. $[Co(NH_3)_6]^{3+}$; 475 nm

B. $[Co(CN)_6]^{3-}$; 310 nm

C. $[Cu(H_2O)_4]^{2+}$; 600 nm

D. $[Ti(H_2O)_6]^{3+}$; 498 nm

Order of λ = C > D > A > B

(62) **Answer :** (3)
Solution:

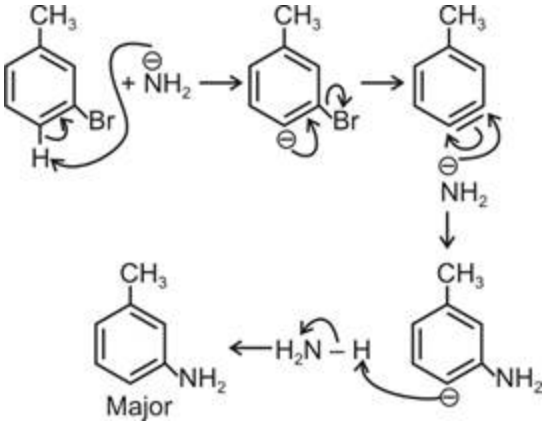
$[Co(NH_3)_3(NO_2)_3]$ will not show optical isomerism but $[Co(C_2O_4)_3]^{3-}$ will show optical isomerism.

$[Co(NH_3)_3(NO_2)_3]$ forms fac-mer isomers.

(63) **Answer :** (2)
Hint:
Reaction follows benzyne mechanism.

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Solution:

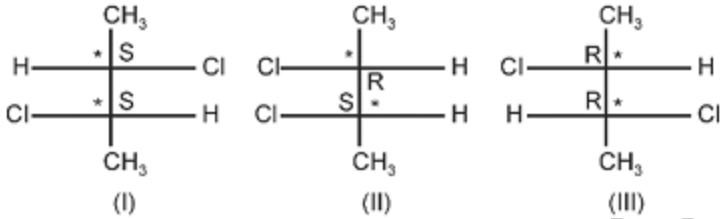


(64) **Answer :** (4)

Hint:

In order to convert a molecule with two stereogenic centers to its enantiomer, the configuration at both centres must be reversed. Reversing the configuration at only one stereogenic centre converts it to a diastereomeric structure.

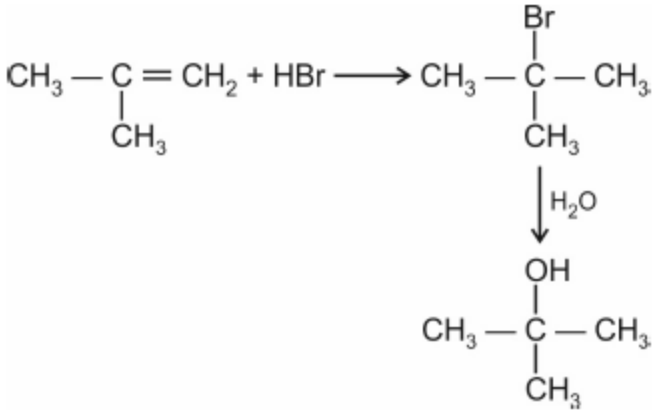
Solution:



(65) **Answer :** (4)

Solution:

HBr will show Markovnikov addition.

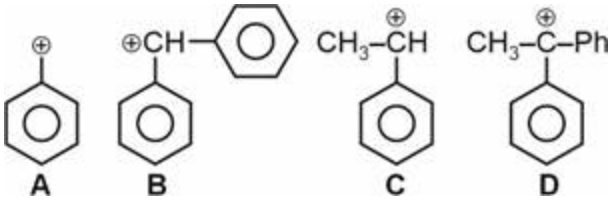


(66) **Answer :** (3)

Hint:

Rate of S_N1 reaction depends upon the stability of carbocation.

Solution:



Order of stability of carbocation is

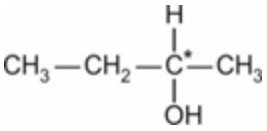
D > B > C > A

(67) **Answer :** (2)

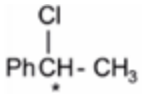
Hint:

Chiral compound is optically active

Solution:
Butan-2-ol has one chiral carbon hence, optically active.

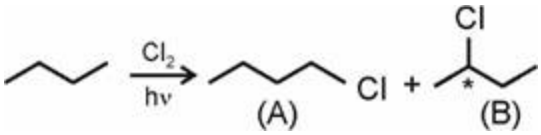


(68) **Answer :** (2)
Solution:
In S_N1 reaction, racemization takes place.



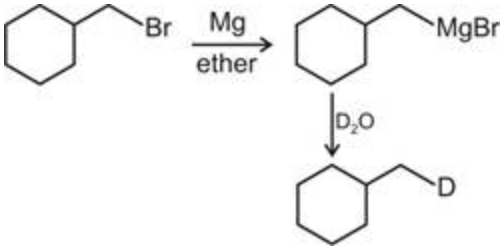
due to having chiral centre as well as stable carbocation shows racemisation.

(69) **Answer :** (1)
Hint:
After monochlorination both structural and stereoisomers should be considered.
Solution:

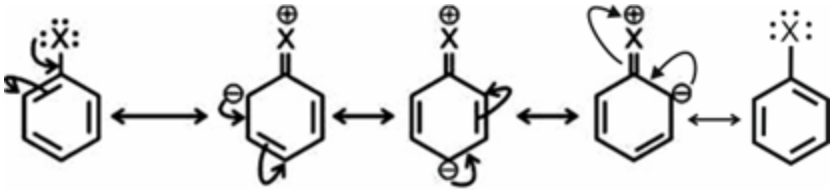


Asterisk carbon represent chiral centre. B will result in a pair of enantiomers.
Total no. of monochlorinated products = 1 + 2 = 3

(70) **Answer :** (3)
Hint:
Alkylhalide on reaction with Mg in presence of ether forms Grignard reagent.
Solution:



(71) **Answer :** (3)
Hint:
In haloarenes, C–X bond acquires a partial double bond character due to resonance.
Solution:

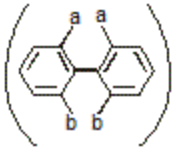


(where X = halogen atom)
A bond cleavage in haloarenes is difficult than haloalkanes and therefore, they are less reactive towards nucleophilic substitution reactions.

(72) **Answer :** (1)
Hint:
Meso compound has plane of symmetry.
Solution:
Meso compounds are optically inactive with chiral carbon.

(73) **Answer :** (4)
Hint:
Biphenyls of type

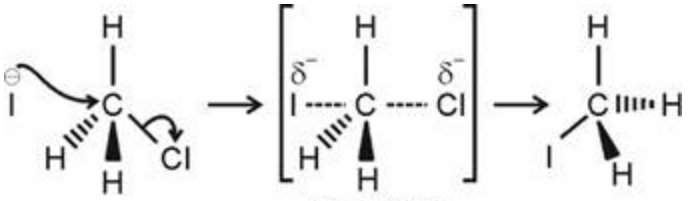
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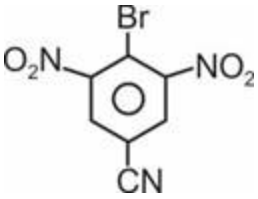
are optically active when a and b are bulky groups

(74) **Answer :** (3)
Solution:
Chloroform is placed in dark coloured bottle and filled till top to avoid its oxidation to toxic phosgene gas.

(75) **Answer :** (2)
Hint:
Least hindered alkyl chloride will react with a very fast rate.
Solution:
CH_3 – Cl is least hindered among the given options.



(76) **Answer :** (4)
Hint:
Compound having less electron density in the ring will give fastest nucleophilic substitution reaction.
Solution:

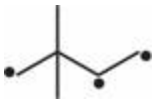


has least electron density in the ring and gives fastest nucleophilic substitution reaction.

(77) **Answer :** (4)
Solution:
Hint: Alkane which has maximum number of different types of hydrogen atoms will give maximum number of monochloro products.
Sol.:
•



(3 monochloro isomers are obtained)
•



(3 monochloro isomers are obtained)
•

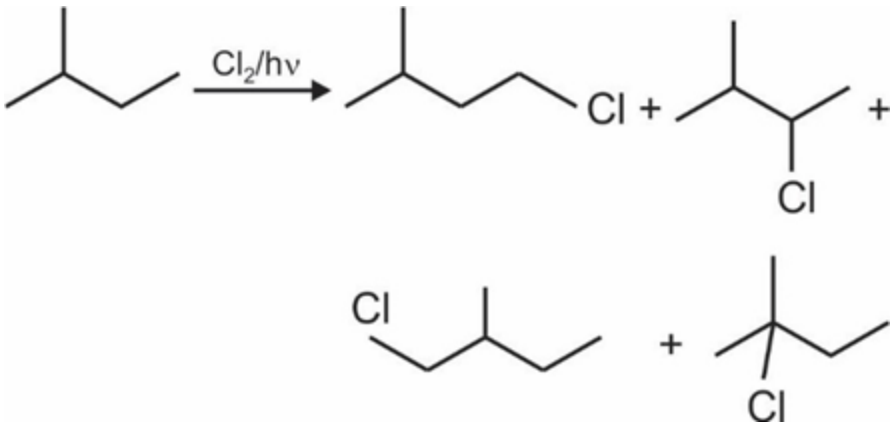


(Only 1 monochloro product is obtained)
•

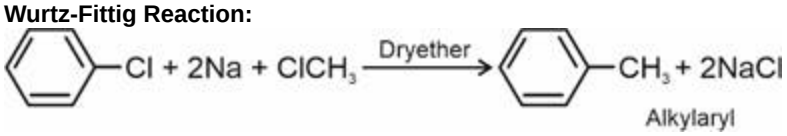


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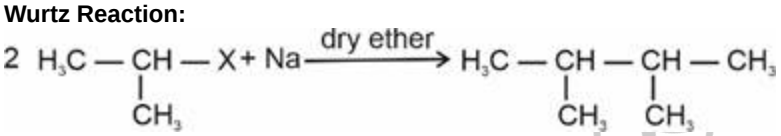
(4 monochloro isomers are obtained)



(78) **Answer :** (1)
Hint:
Wurtz-Fittig Reaction:



Solution:
Wurtz Reaction:



(79) **Answer :** (4)
Hint:
Due to partial double bond character of 'C – Cl' bond in haloarenes, it shows slowest rate towards nucleophilic substitution reaction.

(80) **Answer :** (4)
Hint:
AgCN is a covalent compound and nitrogen is free to donate electron pair.
Solution:
In AgCN, nitrogen atom is free to donate electron pair forming isocyanide as the main product while reacting with alkyl halide
R – X + AgCN → RNC + Ag

(81) **Answer :** (1)
Solution:
$[Co(NH_3)_5Cl]Cl \Rightarrow [Co(NH_3)_5Cl]^{+} + Cl^{-} \Rightarrow Co^{2+}$

(82) **Answer :** (4)
Solution:
$[Pt(NH_3)_6]Cl_4$ has highest conductance because there are 4 Cl^{-} ions in the primary valency and they are responsible for the enhanced molar conductance.

(83) **Answer :** (2)
Solution:
Blue vitriol $\Rightarrow CuSO_4 \cdot 5H_2O$
In this structure, there is 1 hydrogen bonded water molecule.

(84) **Answer :** (2)
Solution:
$[Ni(CN)_4]^{2-} \Rightarrow 3d^8 \Rightarrow$

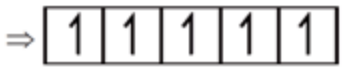


$\Rightarrow dsp^2$

⇒Square planar

(85) **Answer :** (3)
Solution:

$[FeF_6]^{3-} \Rightarrow Fe^{3+} \Rightarrow 3d^5$

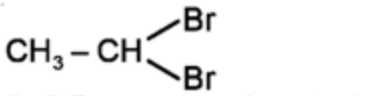


(∴ F^- is a weak field ligand)

⇒ Five unpaired electrons

$\Rightarrow \mu = \sqrt{5(5+2)}\ \text{BM} = \sqrt{35}\ \text{BM} = 5.92\ \text{BM}$

(86) **Answer :** (1)
Solution:



(both Br on same carbon atom)

(87) **Answer :** (3)
Solution:



(88) **Answer :** (4)
Solution:
Dipole moment order is
$CH_3Cl > CH_3F > CH_3Br > CH_3I$

(89) **Answer :** (4)
Solution:
Swarts reaction is used for the synthesis of alkyl fluorides.

(90) **Answer :** (4)
Solution:

$$R\text{–}OH \xrightarrow[\left(\text{Darzen 's method}\right)]{SOCl_2} RCl + SO_2 + HCl$$

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BOTANY

(91) **Answer :** (4)
Solution:
For evolution and speciation, polymorphism plays very important role.
If an inheritable mutation is observed in a population at high frequency, it is referred to as DNA polymorphism.

(92) **Answer :** (3)
Solution:
DNA has evolved from RNA with chemical modifications that makes it more stable.

(93) **Answer :** (4)
Solution:
Transformation occurs when one bacterium picks up free-floating DNA and incorporates it into its own genome.

(94) **Answer :** (4)
Solution:
RNA polymerase II catalyses the transcription of hnRNA. σ-factor facilitates the initiation of transcription in prokaryotes.
Release factor facilitates the termination of translation. 2D structure of tRNA is clover leaf-like.

(95) **Answer :** (1)
Solution:
Marshall Nirenberg is best known for his contribution to the development of a cell-free system for protein synthesis.

(96) **Answer :** (2)
Solution:
The blind approach of simply sequencing the whole set of genome that contains all the coding and non-coding sequence, and later assigning different regions in the sequence with functions (a term referred to as Sequence annotation). The EST is referred to as identifying all the genes that are expressed as RNA.

(97) **Answer :** (1)
Solution:
In Meselson and Stahl experiment, density gradient bands was formed by using caesium chloride.

(98) **Answer :** (4)
Solution:
For the codon 5′ UAC 3′, the sequence of anticodon on tRNA is 3′ AUG 5′

(99) **Answer :** (1)
Solution:
Prokaryotes lack primary transcript. Nuclear and cytoplasmic RNA polymerases are collectively more than three in number. Splicing machinery evolved around RNA but splicing is not commonly observed in prokaryotes. Tailing occurs in template independent manner.

(100) **Answer :** (2)
Solution:
Hershey and Chase worked with T_2 bacteriophage and *E*. *coli* to prove that DNA acts as genetic material. Taylor *et*. *al* had proved semi-conservative mode of chromosome replication in root of *Vicia faba*. Avery MacLeod and McCarty performed biochemical characterisation of transformation principle, *in-vitro*.

(101) **Answer :** (3)
Hint:
Number of phosphodiester bonds depends on the circular or linear nature of genetic material.
Solution:
Number of phosphodiester bonds in a linear dsDNA molecule = number of total nucleotides – 2.

(102) **Answer :** (1)
Hint:
lac z, *y* and *a* are structural genes in *lac* operon.
Solution:
Repressor protein is coded by inhibitor gene *i*. It is not a structural gene.

(103) **Answer :** (3)
Solution:
Genetic code is degenerate because some amino acids are coded by more than one codons. Given statement does not provide any idea about degeneracy of codons.

(104) **Answer :** (2)
Solution:
SSB proteins or ‘Single Strand Binding’ protein bind and stabilise the single stranded DNA during DNA replication.

(105) **Answer :** (2)
Solution:
Deoxyadenosine is a nucleoside, which lacks phosphate group.

(106) **Answer :** (3)
Solution:
Friedrich Meischer named the acidic substance present in nucleus as Nuclein.

(107) **Answer :** (1)
Solution:
If the sequence of bases in one strand is known, then the sequence in other strand can be predicted because both the strands of DNA are complementary to each other.

(108) **Answer :** (1)
Hint:
hnRNA undergoes splicing to form mRNA.
Solution:
Transcription produces hnRNA that requires splicing to form mRNA. Only replication uses both strands of DNA as template.

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(109) Answer : (3)
Solution:
If the sequence of mRNA is 5' CACUGCAUGUGGGAG 3' then the sequence of the template strand will be 3' GTGACGTACACCCTC 5'

(110) Answer : (1)
Solution:
If a piece of DNA is needed to be propagated during recombinant DNA procedures, it requires a vector because of the requirement of the origin of replication which will be provided by the vector inside the *E.coli*

(111) Answer : (4)
Hint:
Transcription is copying genetic information from one strand of the DNA into RNA.
Solution:
The DNA sequence coding for tRNA or rRNA molecules also define a gene.

(112) Answer : (3)
Hint:
tRNA is also known as the adapter molecule.
Solution:
All tRNA molecules have a guanine residue at its 5' terminal end. At 3'-end, unpaired -CCA sequence is present.

(113) Answer : (3)
Solution:
Inducer is not a component of transcription unit.

(114) Answer : (1)
Solution:
The discovery of nuclein by Meischer and the proposition for principles of inheritance by Mendel were almost at the same time, but that the DNA acts as a genetic material took long to be discovered and proven. By 1926, the quest to determine the mechanism for genetic inheritance had reached the molecular level.

(115) Answer : (3)
Solution:
Peptidyl transferase catalyses the formation of peptide bond.

(116) Answer : (4)
Hint:
RNA has same sequence as coding strand, except thymine at the place of uracil.
Solution:
Coding strand → 5′ – GACCCAAGCTACGG –3′
RNA → 5′ – GACCCAAGCUACGG – 3′

(117) Answer : (4)
Solution:
The human genome contains 3164.7 million bp. The average gene consists of 3000 bases, but sizes vary greatly. Repetitive sequences are thought to have no direct coding functions, but they shed light on chromosome structure, dynamics and evolution.

(118) Answer : (4)
Solution:
The last step of DNA fingerprinting is detection of hybridised DNA fragments by autoradiography.

(119) Answer : (2)
Solution:
Francois Jacob and Jacque Monod proposed a model of gene regulation, known as operon model in bacteria.

(120) Answer : (4)
Solution:
Genetic code is non-overlapping in nature.

(121) Answer : (2)
Solution:
Some amino acids are coded by more than one codon hence the code is degenerate.

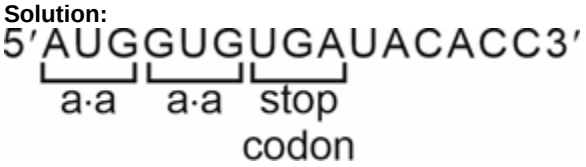
(122) Answer : (1)
Solution:
Regulation of gene expression is not exerted at replication level.

(123) Answer : (4)

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Solution:
Chargaff's rule is not applicable for single stranded DNA.

(124) Answer : (1)
Solution:



(125) Answer : (3)
Solution:
Euchromatin is loosely packed, transcriptionally active region of chromatin.

(126) Answer : (3)
Solution:
Translation is synthesis of polypeptide according to codons present on m-RNA which does not follow rule of complementarity.

(127) Answer : (3)
Solution:
Lactose acts as an inducer. Regulation of lac operon by repressor protein is negative regulation. When lactose is absent repressor binds to the operator and in the presence of lactose repressor binds to lactose. Lac operon is inducible operon system.

(128) Answer : (3)
Solution:
In dsDNA, purines of one strand are paired with pyrimidines of corresponding strand by formation of hydrogen bonds. Adenine forms two H-bonds with thymine and guanine is bonded with cytosine with three H-bonds.

(129) Answer : (2)
Solution:
Sequencing completed in the year 2006

(130) Answer : (1)
Solution:
Caenorhabiditis elegna, Drosophila

(131) Answer : (2)
Solution:
Activation of amino acids during translation is done by Aminoacyl-tRNA synthetase.

(132) Answer : (4)
Solution:
If repressor is attached at operator their is no synthesis of mRNA.

(133) Answer : (3)
Solution:
Movement of ribosome on mRNA is called translocation

(134) Answer : (1)
Solution:
ϕ × 174 bacteriophage – 5386 nucleotides, Bacteriophage lambda – 48502 bp
E. coli – 4.6×10^6 bp
Human genome – 3.3×10^9 bp

(135) Answer : (4)
Solution:
mRNA of eukaryotes carry poly A tail during post-transcriptional change.

ZOOLOGY

(136) Answer : (4)
Solution:
Homo sapiens evolved in Africa.

(137) Answer : (1)
Solution:
Flippers of Penguins and Dolphins are an example of analogous organs. Analogous structures are a result of convergent evolution.

(138) Answer : (3)
Solution:
Australian marsupials and placental mammals show convergent evolution.

(139) Answer : (3)
Solution:
Around 200 mya, some of the land reptiles went back in to water and evolved into fish-like reptiles (*Ichthyosaurs*)

(140) Answer : (3)
Solution:
Evolution is not always from simple form to complex form.

(141) Answer : (2)
Solution:
The chronological order of human evolution from early to recent ones :
Ramapithecus → *Australopithecus* → *Homo habilis* → *Homo erectus* → Neanderthal man → Modern *Homo sapiens*

(142) Answer : (3)
Solution:
Bryophytes arose from chlorophyte ancestors. Monocots evolved after dicots.

(143) Answer : (2)
Solution:
The Neanderthal man had a cranial capacity of 1400cc.

(144) Answer : (3)
Solution:
Stabilising natural selection is a type of selection that favors the average individuals in a population. It reduces variation and maintains the status quo for a particular trait. In this context, the option that represents stabilising natural selection is: Average body weight at which newborn baby is more likely to survive
This indicates that individuals with average body weight have a higher survival rate, which is a characteristic of stabilising selection.

(145) Answer : (4)
Solution:
Unit of evolution is population while unit of natural selection is individual. Bryophytes were originated from chlorophyte – ancestors in carboniferous period.
The brain capacities of *Homo habilis* were between 650–800cc. They probably did not eat meat. Fossils of *Homo erectus* were discovered in Java in 1891.

(146) Answer : (2)
Solution:
Continental drift created a biogeographical barrier for placentals.

(147) Answer : (1)
Solution:
Excess use of herbicides, pesticides, *etc*., has only resulted in selection of resistant varieties in a much lesser time scale. This is also true for microbes against which we employ antibiotics or drugs against eukaryotic organisms/cells. Hence, resistant organisms/cells are appearing in a time scale of months or years and not centuries. These are examples of evolution by anthropogenic actions.

(148) Answer : (4)
Hint:
Embryology.
Solution:
On careful study performed by Ernst von Baer, the theory given by Ernst Haeckel i.e., Embryological support for evolution was disapproved which stated that certain features during embryonic stage are common to all vertebrates, that may be absent in adult. Reproductive fitness was a concept given by Darwin.

(149) Answer : (2)
Solution:
Sweet potato (root modification) and potato (stem modification) are examples of analogous structure.
Analogous structures are the result of convergent evolution.

(150) Answer : (1)

Solution:
Microscopic microbes have shorter lifespans, so they divide faster and become millions of individuals within hours. Hence, microbes have more chances to evolve faster.

(151) Answer : (1)
Solution:
All the existing life forms share similarities and share common ancestors. However, these ancestors were present at different periods in the history of Earth (epochs, periods and eras). The geological history of Earth closely correlates with the biological history of Earth.

(152) Answer : (4)
Solution:
Early Greek thinkers thought units of life called spores were transferred to different planets including Earth. 'Panspermia' is still a favourite idea for some astronomers.

(153) Answer : (3)
Solution:
Louis Pasteur by careful experimentation demonstrated that life comes only from pre-existing life and disproved spontaneous generation theory.

(154) Answer : (2)
Solution:
The essence of Darwinian theory of evolution is natural selection. If changes in allele frequency occurs by chance, it is called genetic drift. Theory of use and disuse of organs was proposed by Lamarck.

(155) Answer : (2)
Solution:
In Hardy-Weinberg equation, p^2 represents homozygous dominant, q^2 represents homozygous recessive and 2pq represents heterozygous individuals.

(156) Answer : (4)
Solution:
Speciation through reproductive isolation is one of the important consequences of geographical isolation.

(157) Answer : (4)
Solution:
Five factors are known to affect Hardy-Weinberg equilibrium. These are gene migration or gene flow, genetic drift, mutation, genetic recombination and natural selection.

(158) Answer : (2)
Solution:
Correct sequence of geological time scale is
Eras – Periods – Epochs

(159) Answer : (1)
Hint:
Sulphur containing compound.
Solution:
S. L. Miller created electric discharge in a closed flask containing CH_4, H_2, NH_3 in 2 : 2 : 1 ratio and water vapour at 800°C.

(160) Answer : (3)
Hint:
Raisen district
Solution:
Cave paintings by pre-historic humans can be seen at Bhimbetka rock shelter in Raisen district of Madhya Pradesh.

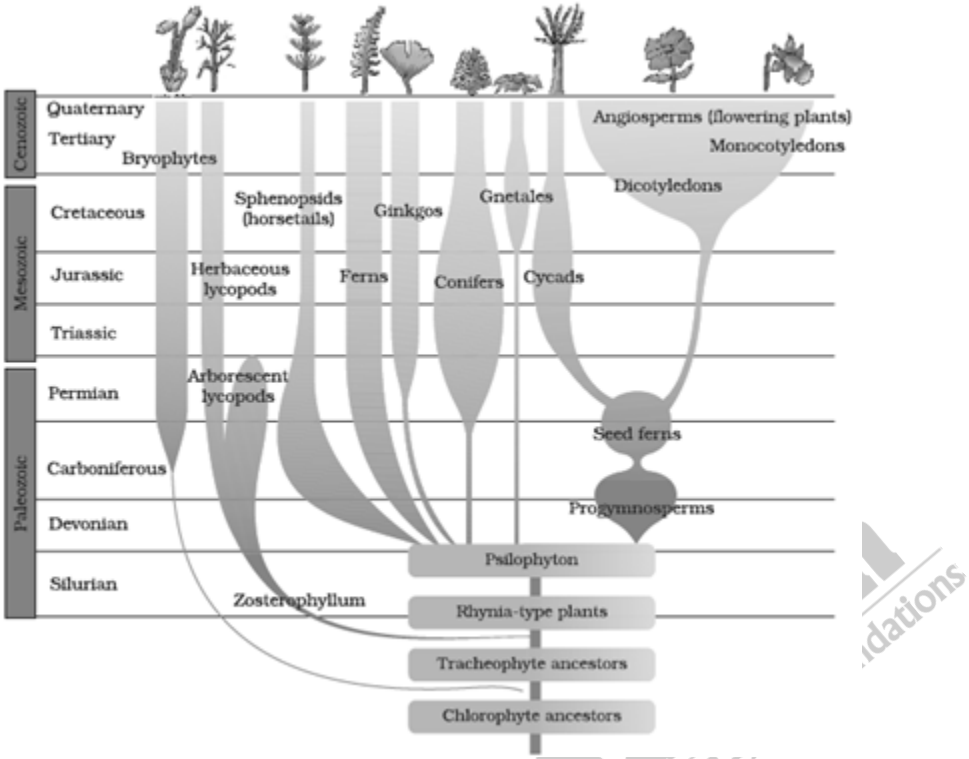
(161) Answer : (1)
Hint:
Hardy Weinberg equilibrium.
Solution:
Frequency of recessive allele *i.e.* q = 0.8; Frequency of dominant allele *i.e.* p = 1 – 0.8 = 0.2
$\therefore$ Frequency of homozygous dominant individuals (AA) = $p^2 = (0.2)^2 = 0.04$
Frequency of heterozygous individual (Aa) = 2pq = 2(0.2) (0.8) = 0.32
Frequency of homozygous recessive individual (aa) = $q^2 = (0.8)^2 = 0.64$

(162) Answer : (4)
Hint:
Brachiosaurus had long neck and long tail.
Solution:

Statement B is incorrect because *Stegosaurus* had big kite like plates on its back for protection.

(163) Answer : (2)
Solution:
Two key concept of Lamarckism are
(i) Inheritance of acquired characters and
(ii) Use and disuse theory. According to Lamarck, the changes acquired by the organisms during their life time are passed on the next generation and development of an organ is directly proportional to its use.

(164) Answer : (1)
Solution:



(165) Answer : (3)
Solution:
Flying squirrel and bobcat are placental mammals that show homology.

(166) Answer : (3)
Solution:
Analogous structures are not anatomically similar structures though they perform similar functions. Hence, analogous structures are the result of convergent evolution.

(167) Answer : (2)
Solution:
Charles Darwin implied natural selection as a mechanism of evolution. The fitness according to Darwin, refers ultimately and only to reproductive fitness. Hence, those who are better fit in an environment have more progeny than others. Thus, therefore, will survive more and hence are selected by nature. He called it natural selection. Ernst Heckel gave embryological support for evolution, Hugo de Vries proposed mutation theory and Lamarck proposed theory of use and disuse of organs.

(168) Answer : (3)
Solution:
Louis Pasteur by his experiment using a swan necked flask and killed yeast finally dismissed the theory of abiogenesis.

(169) Answer : (1)
Solution:
In the solar system of the Milky way galaxy, Earth was supposed to have been formed about 4.5 billion years back. The universe is very old-almost 13.8 billion years old.

(170) Answer : (3)
Solution:
Industrial melanism occurred due to human activity resulting in increased industrialisation. The darker variety of moth was existing prior to pollution and arose possibly due to random mutations.

(171) **Answer :** (1)
Solution:
Chemoheterotrophs depend on organic chemicals to obtain energy.

(172) **Answer :** (3)
Solution:
Predator is the agent of natural selection in this case.

(173) **Answer :** (1)
Solution:
Presence of scales on hind limbs and shelled/cleidoic eggs are considered as reptilian ancestry of birds because both features together appeared first in reptiles but are also found in birds. Birds lack teeth and have a four chambered heart

(174) **Answer :** (1)
Solution:
According to Ernst Haeckel, it is summarised as biogenetic law. Ontogeny is development of embryo and phylogeny is the ancestral sequence. Biogenetic law states that ontogeny is recapitulation of phylogeny. As per this law, the sequence of embryonic development in different vertebrates show striking similarities.

(175) **Answer :** (1)
Solution:
The most probable sequence in case of human evolution is as follows:
Ramapithecus → Australopithecus → Homo habilis → Homo erectus → Homo sapiens

(176) **Answer :** (3)
Solution:
The fossils of *Homo erectus* was found in Java in 1891.

(177) **Answer :** (2)
Solution:
Connecting links show characteristics of both the divisions.

(178) **Answer :** (2)
Solution:
Age of fishes = Devonion period.
Age of mammals = Cenozoic era.

(179) **Answer :** (2)
Solution:
Theory of spontaneous generation or abiogenesis states that life has originated from non-living matter.

(180) **Answer :** (3)
Solution:
UV rays broke H_2O into H_2 and O_2.

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